1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L16 – Rapid Prototype Evaluation II**

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3 **Recap**

- We discussed expert evaluation
- Today we shall discuss about another way to rapidly evaluate prototypes

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User Evaluation

5 **Basic Idea**

- With expert evaluation, we rely on data provided by those who are not likely to be the users of the system
  - All right in early phases
- We are likely to get more insight from data collected from the users themselves

6 **Basic Idea**

- One way to do that is to conduct empirical studies – shall learn about it in more details later
- Involves elaborate setup and expertise to collect and analyze data

7 **Basic Idea**

- There is another (much less complex) method – get feedback from users by asking them questions directly – called evaluation with self-reports

8 **Basic Idea**

- Data collected in this method sometimes called subjective or preference data

9

Evaluation with Self Reports

10 **Basic Idea**

- Many ways to collect user feedback
  - Rating on a scale (e.g. Likert scale)
  - Can ask them to choose from a list of system attributes
  - Ask direct, open-ended questions (e.g., "what you feel about the system?"

11 **Basic Idea**

- Problems with open-ended questions –feedback may not be *reliable*
- Feedback on rating scales are generally considered more efficient and reliable

12 **Basic Idea**

- Collection of feedback with rating scales is not easy
- TWO issues

13 **Issue 1 with Feedback**

- We collect ratings on *items* – questions or statements
  - E.g. - rate the statement "The controls are easily accessible" on a 3-point Likert scale
  - Pre-requisite - identify items (commonly referred to as a *questionnaire* although it need not contain questions)
- Challenging to identify suitable questionnaire

14 **Issue 2 with Feedback**

- When we ask for feedback, we assume user is already exposed to the system
  - S/he might be asked to use the system before feedback, like in a controlled study
  - In such studies, users are typically asked to perform a set of tasks

15 **Issue 2 with Feedback**

- Thus, we have two points for data collection
  - After every task
  - At the end of the session (after all the tasks)
- Desirable to collect data at both the points
- Corresponding questionnaire may not be the same

16 **Post-Session vs Post-Data Feedback**

- If we collect feedback after each task, we are likely to pinpoint problems w.r.t. specific tasks and parts of the interface used
  - Users are likely to remember problems better just after each task
- However, this approach may not be possible always

17 **Post-Session vs Post-Data Feedback**

- With post-session feedback, we are more likely to learn about user perception on overall system
- It may not be possible to relate that perception with specific task or components of the interface

18 **Questionnaire (Some)**

19

Example of Post-Task Questionnaire - ASQ

20 **Basic Idea**

- Consists of THREE statements
- Users asked to provide rating on a 7-point rating scale for each statement
  - Ratings range from "strongly disagree" (rating of 1) to "strongly agree" (the highest rating of 7)

21 **Basic Idea**

22 **Basic Idea**

- Simplicity - designed to collect quick feedback, as feedback is supposed to be collected multiple times

23 **Basic Idea**

- First statement related to effectiveness measure
- Efficiency aspect covered in second statement
- All the three taken together represents satisfaction

24 **Basic Idea**

- Use of the feedback
  - By considering average rating for each statement
  - Or average rating for all statements taken together

25

Example of Post-Session Questionnaire - SUS

26 **Basic Idea**

- Consists of TEN statements
  - Half positively worded (1, 3, 5, 7, 9)
  - Other half negatively worded (2, 4, 6, 8, 10)

27 **Basic Idea**

28 **Basic Idea**

- Rating collected for each statement, on a 5-point "agreement" scale (a Likert scale where 1 = "strongly disagree" and 5 = "strongly agree")

29 **Basic Idea**

- We need to combine the ratings for all the ten statements to obtain an overall SUS score (S)
- Tricky affair

30 **Combining Ratings**

- Sum up *contribution* (C) for each statement to get overall score S
- Contribution from each statement is an integer and lies between 0 and 4
- Two components to calculate individual contributions

31 **Combining Ratings**

- For positively worded statements, individual contribution for each statement is calculated as the rating obtained minus 1
- For negatively worded statements, the contribution is calculated as 5 minus the rating

32 **Combining Ratings**

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33 **Example – How to Combine**

- Statement 1: Rating 5
- Statement 2: Rating 4
- Statement 3: Rating 2
- Statement 4: Rating 1
- Statement 5: Rating 2
- Statement 6: Rating 3
- Statement 7: Rating 2
- Statement 8: Rating 4
- Statement 9: Rating 5
- Statement 10: Rating 2

34 **Example – How to Combine**

- Combining ratings for positively-worded statement

(5-1) + (2-1) + (2-1) + (2-1) + (5-1) = 11

- Combining for negatively-worded statements

(5-4) + (5-1) + (5-3) + (5-4) + (5-2) = 11

- Sum two components (11 + 11) = 22
- Overall score 22×2.5 = 55

35 **Example – How to Combine**

- We can interpret SUS score as percentage
- Thus, we can say score in the example is 55 percent
- It is generally assumed that an SUS score around 70 percent or higher is desirable
- Scores lower than that might indicate usability problems

36 **Book**

- Bhattacharya, S. (July, 2019). Human-Computer Interaction: User-Centric Computing for Design, McGraw-Hill India
  - Print Edition: ISBN-13: 978-93-5316-804-9; ISBN-10: 93-5316-804-X
  - E-book Edition: ISBN-13: 978-93-5316-805-6; ISBN-10: 93-5316-805-8

Chapter 9, Sec 9.3